DS 301 · FINAL PROJECT

Credit Card Default Prediction

Based on Ly, Schuller, Simanic & Singh

Simon · Gabriel · Eduardo · Maria

The article compares three supervised models

Research questions

Which hyperparameters matter?
Which model predicts default best?

Models tested

Decision Tree
KNN
SVM

The article reported SVM as the strongest model, but default-class F1 remained low.

Cleaning leaves 29,601 imbalanced records

The article uses payment, credit, and demographic data from clients in Taiwan.

29,601

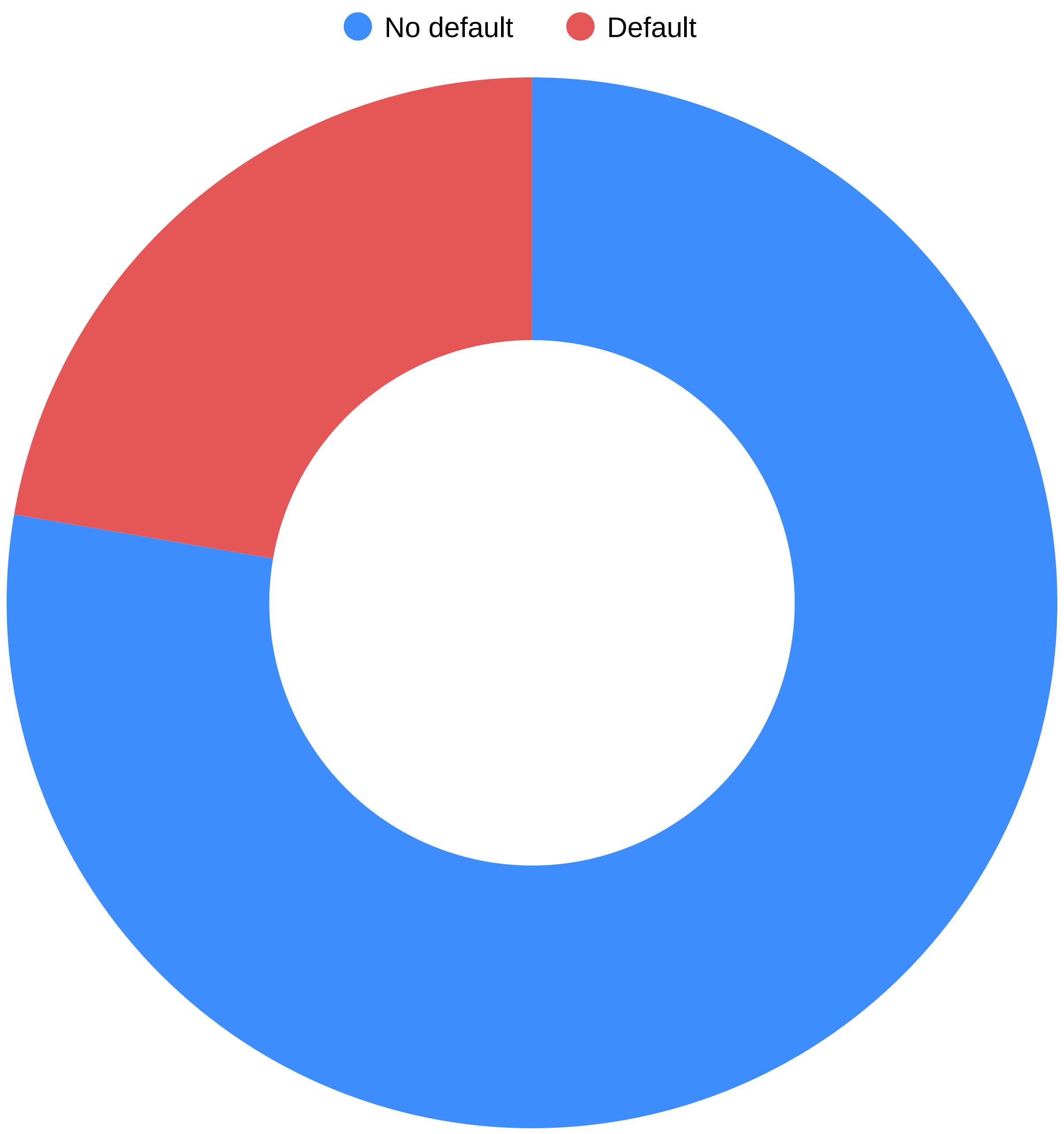
customers

23

input features

22.3%

customers in default



We reproduced the article's workflow

01

Clean

Removed 399 rows with undocumented categories.

02

Split

Used 80% for training and 20% for testing.

03

Prepare

Standardized and oversampled the training set.

04

Reproduce

Reported best parameters
DT · KNN · SVM

Reproduced models + Logistic Regression

Article reproduction

Decision Tree · KNN · SVM
Reported best parameters
Same cleaned and balanced data

Group contribution

Added Logistic Regression
Tuned regularization C
Selected with default-class F1
Evaluated on the same test set

One controlled comparison: the article models and our added model use the same cleaned data and test set.

Logistic Regression: our fourth model

Why we added it

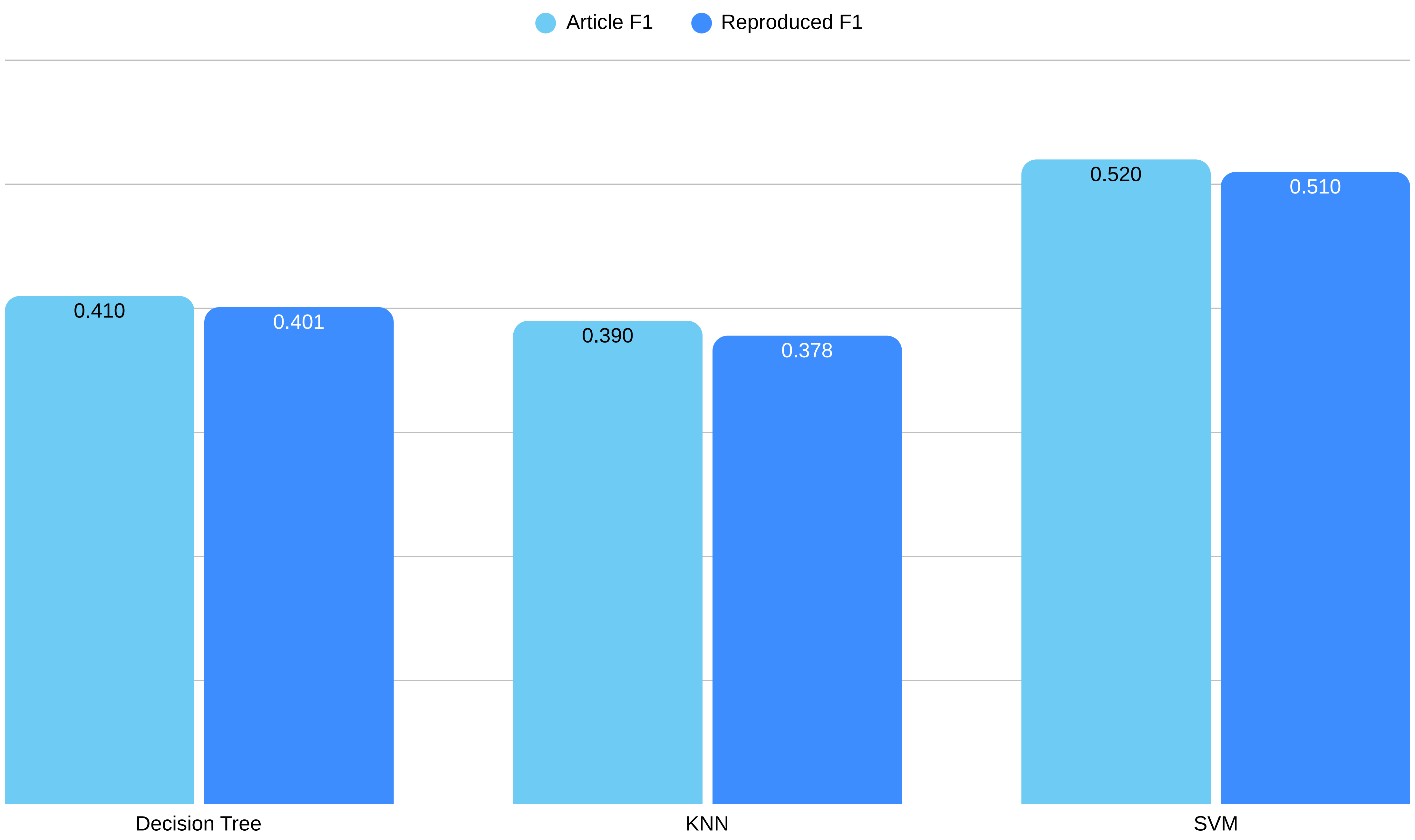
Binary classification baseline
Easy to explain
Produces default probabilities
Fast to train

How we implemented it

Same preprocessed data
C: 0.01, 0.1, 1, 10, 100
Solver: liblinear
Selected with F1
Best C: 0.01

Accuracy 0.696 · Precision 0.390
Recall 0.645 · F1 0.486

SVM led F1; Logistic led recall



Article winner

SVM
F1-score 0.520

Reproduced winner

SVM
F1-score 0.510

Group contribution

Logistic Regression
Recall 0.645 · F1 0.486

Four challenges shaped our evaluation

01 **Class imbalance**

Only 22.3% default, so accuracy can mislead.

02 **Exact reproduction**

The paper does not provide its random seed.

03 **Computation cost**

SVM required much more training time.

04 **Business decision**

Higher recall can create more false positives.

Four practical lessons from the project

01 Metrics

Accuracy alone is incomplete for imbalanced data.

02 Preprocessing

Cleaning, scaling, and oversampling shaped the results.

03 Reproduction

Results can vary when seeds or details are missing.

04 Business context

The cost of errors should guide model selection.

SVM won overall.
Logistic Regression added insight.

Its recall of 0.645 detected the largest share of default clients, showing why the best model depends on the business objective.

Q&A

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Thank You

Simon · Gabriel · Eduardo · Maria